Exact Consecutive-Pattern Popularity in the Class Avoiding 123, 231 and 312

Ryutaro Yonezu

Independent Researcher | Preprint v1.0 | 5 November 2026

Abstract

For permutations avoiding the consecutive patterns 123, 231 and 312, we identify the total number of consecutive 321 occurrences, counted over all permutations and all positions, as $E_{n+1} - 2E_n$ for every $n \geq 3$, where E_n are the Euler zigzag numbers. Reverse-complement symmetry gives exact totals for 132 and 213. With the normalization of Hassler and Kirgizov, the limiting popularities are $4/\pi^2$ for 321 and $1/2 - 2/\pi^2$ for each of 132 and 213. The argument marks an edge in the known circular-difference integral-operator model. The avoidance count, Euler-number sequence, and underlying operator are prior work. This is a limited preprint for one consecutive-avoidance class; novelty remains **INCONCLUSIVE**, and human peer review has **not been performed**.

Audit and publication disclosures

The supplied Claude Sonnet 5.5 report records v0.2 MODEL / PROOF / COMPUTATION PASS and v0.3 differential audit PASS. The reviewer did not inspect HK v2, EKP Section 7 or Zhao (2014), and did not execute Windows. ChatGPT's primary-source checks are attributed separately. Windows v0.4.1 reproduction PASS is supported by the owner's supplied console log, not by execution in this publication session or by Claude. The v0.4 mathematics and code remain fixed; historical HOLD records remain archived. Human peer review has **not been performed**.
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1 Definitions and claimed theorem

Let $A_n = \mathrm{Av}_n(123, 231, 312)$ be the set of length-n permutations avoiding the three indicated **consecutive** patterns. Let E_n be the Euler up/down (zigzag) numbers characterized by

$A(z) = \sum_{n\geq 0} E_n z^n/n! = \sec z + \tan z$.

Let $N_n(p)$ be the sum of the numbers of consecutive occurrences of pattern p over **all** permutations in A_n, counting each location separately.

Theorem (independent AI mathematical audit reported PASS; novelty not established). For every integer $n \geq 3$,

1. $|A_n| = nE_{n-1}$ (known, included only as a cross-check).
2. $N_n(321) = E_{n+1} - 2E_n$.
3. $N_n(132) = N_n(213) = \frac{(n-2)nE_{n-1} - E_{n+1} + 2E_n}{2}$.
4. Using the original paper's normalization $\mathrm{pop}(p) = \lim_{n\to\infty} N_n(p)/(n|A_n|)$, whenever the limit is evaluated: $\mathrm{pop}(321) = 4/\pi^2$, $\mathrm{pop}(132) = \mathrm{pop}(213) = 1/2 - 2/\pi^2$.

Since π is transcendental, the three limits above are irrational (indeed transcendental, since a nonzero rational affine transform of a transcendental number is transcendental).

2 Uniform real variables and circular differences

Let $x_1, \ldots, x_n$ be independent Unif$(0,1)$ random variables. Almost surely they are distinct, and their rank permutation is uniformly distributed on S_n. Define

$u_j = \{x_j - x_{j+1}\}$, for $1 \leq j \leq n-1$,

where braces denote fractional part in $[0,1)$.

The map $(x_1, \ldots, x_n) \longmapsto (x_1, u_1, \ldots, u_{n-1})$ is an invertible triangular group transformation on the torus modulo one, preserving normalized Lebesgue measure. Therefore the anchor x_1 and all u_j are mutually independent Unif$(0,1)$ random variables. Conditional on the entire u vector, each x_j is uniform on $[0,1)$.

Local equivalence: for any triple of distinct reals x_j, x_{j+1}, x_{j+2}, the associated rank pattern is in $\{132, 213, 321\}$ if and only if $u_j + u_{j+1} < 1$. The other three patterns $\{123, 231, 312\}$ correspond to $u_j + u_{j+1} > 1$. Equality occurs only on a measure-zero set.

Indeed, the three positions on the circle are x_j, $\{x_j - u_j\}$, and $\{x_j - u_j - u_{j+1}\}$. When $u_j + u_{j+1} < 1$, the rank pattern is 321 precisely when $x_j \geq u_j + u_{j+1}$, a conditional event of probability $1 - u_j - u_{j+1}$.

Write $m = n-1$, $\Omega_m = \{u \in [0,1]^m : u_j + u_{j+1} < 1 \text{ for } j = 1, \ldots, m-1\}$. It follows that

$|A_n|/n! = \int_{\Omega_{n-1}} 1 \, du$, and

$N_n(321)/n! = \int_{\Omega_{n-1}} \left[\sum_{j=1}^{n-2} (1 - u_j - u_{j+1})\right] du$.

Audit checkpoint A: Independently verify the torus measure-preserving change of variables, the six-pattern case split, and the weight $1 - u_j - u_{j+1}$. All later equations depend on this step.

3 Resolvent and Euler EGF

Define the integral operator on continuous functions

$$(Uf)(t) = \int_0^{1-t} f(s)\, ds$$

and its resolvent applied to 1 as a power series

$F(z,t) = \sum_{k\geq 0} z^k (U^k 1)(t)$.

Initially the series converges for $|z| < 1$, which suffices to justify term-by-term integration and differentiation in t locally near $z = 0$. The identity $F = 1 + zUF$ gives

$\frac{\partial F(z,t)}{\partial t} = -zF(z, 1-t)$, $F(z,1) = 1$, and $\frac{\partial^2 F}{\partial t^2} = -z^2 F$.

Solving yields

$F(z,t) = A(z)\cos(zt) - \sin(zt)$,

where $A(z) = (1+\sin z)/\cos z = \sec z + \tan z$, since $F(z,1) = 1$.

For $m \geq 1$, the integral $\int_{\Omega_m} 1\, du = \int_0^1 (U^{m-1}1)(t)\, dt$. Consequently

$\sum_{m\geq 1} z^m \operatorname{vol}(\Omega_m) = z\int_0^1 F(z,t)\, dt = F(z,0) - 1 = A(z) - 1.$

Coefficient comparison yields $\operatorname{vol}(\Omega_m) = E_m/m!$, and hence the known $|A_n| = nE_{n-1}$.

Audit checkpoint B: Check both boundary conditions, the series index $m = n-1$, and the normalizing $n!$.

4 Marking a 321 occurrence and the exact EGF

Set

$J(z) = \int_0^1 \int_0^{1-u} (1-u-v)F(z,u)F(z,v)\, dv\, du.$

Choosing a marked adjacent pair u_j, u_{j+1} in an admissible chain $u_1, \ldots, u_{n-1}$, the chain on either side of that edge contributes powers of U. Thus the product of the two F factors enumerates all left/right extensions. Since the two marked variables contribute z^2, while the original occurrence-count EGF is indexed by permutation length $n = m+1$, we get the **crucial shift**

$H(z) := \sum_{n\geq 3} N_n(321) z^n/n! = z^3 J(z).$

The following algebra evaluates the integral exactly. From $F(z,t) = 1 + z\int_0^{1-t} F(z,s)\, ds$, first obtain

$z\int_0^{1-u} (1-u-v)F(z,v)\, dv = \int_u^1 (F(z,y)-1)\, dy.$

Therefore, interchanging integration order,

$$J(z) = \frac{1}{z}\int_0^1 F(z,u) \int_u^1 (F(z,y)-1)\, dy\, du$$

$$= \frac{1}{z}\int_0^1 (F(z,y)-1) \int_0^y F(z,u)\, du\, dy$$

$= \frac{1}{z^2}\int_0^1 (F(z,y)-1)(F(z,1-y)-1)\, dy.$

Use $F_y(z,y) = -zF(z,1-y)$, $F(z,0) = A(z)$, $F(z,1) = 1$, and $z\int_0^1 F(z,y)\, dy = A(z) - 1$. Then

$$z^3 J(z) = -\int_0^1 (F-1)F_y\, dy - z\int_0^1 (F-1)\, dy$$

$$= (A-1)^2/2 - (A-1) + z$$

$= (A^2 - 4A + 3)/2 + z.$

Finally $A' = (A^2+1)/2$, as follows by differentiating $\sec z + \tan z$. Therefore

$H(z) = A'(z) - 2A(z) + 1 + z.$

The coefficients of z^0, z^1, z^2 are zero. Comparing the coefficient of $z^n/n!$ for every $n \geq 3$ proves

$N_n(321) = E_{n+1} - 2E_n$.

Audit checkpoint C: The most important possible failure points are the z^3 shift in H, swapping integrals, and the last integration by parts. Check each separately, without relying on numerical data.

5 The two other allowed patterns

Reverse-complement $(R \circ C)$ fixes the forbidden set $\{123, 231, 312\}$ (it fixes 123 and swaps 231 and 312), and exchanges the permitted patterns 132 and 213, while fixing 321. Hence $N_n(132) = N_n(213)$ exactly for all n.

Every admissible permutation contributes $n-2$ consecutive triples, all in $\{132, 213, 321\}$; therefore

$N_n(132) + N_n(213) + N_n(321) = (n-2)|A_n|$.

Combining this relation with the established expression for $N_n(321)$ proves claim 3.

6 Asymptotic normalization (important correction)

Hassler–Kirgizov (arXiv:2511.02442v2, their Eq. (1.1)) use $n|A_n|$ as the denominator in their definition of pop. The alternative $(n-2)|A_n|$ is a natural normalization over actual length-three windows but differs at finite n; both have the same limit.

The standard asymptotic of Euler zigzag numbers is

$E_n/n! \sim 2(2/\pi)^{n+1}$.

Hence, using $|A_n| = nE_{n-1}$,

$\frac{N_n(321)}{n|A_n|} = \frac{E_{n+1}-2E_n}{n^2 E_{n-1}} \longrightarrow \frac{4}{\pi^2}$.

The other two limits follow from symmetry and the sum of $(n-2)|A_n|$ windows. This gives both a solution candidate for their open Class 12 and an irrational asymptotic popularity example.

7 Independent arithmetic checking

Run accompanying `No54_v0.1_VERIFY_20261105.py` with standard-library Python 3.10+; it builds Euler numbers using a recurrence and separately brute-forces all permutations for $3 \leq n \leq 9$. It checks the avoidance-set count, each allowed-pattern count, and the exact formulas. Additional values for $n = 10, 20, 40, 60, 100, 120$ are generated by exact integer arithmetic. These computations do **not** replace proof or external review.

8 Prior work and scope of the contribution

Hassler and Kirgizov [1], Table 1, list this exact consecutive-avoidance class as Class 12; their equation (1.1) supplies the denominator $n|A_n|$. Their v2 asks for this class's popularities. The statement addresses this class and length-three consecutive statistics only.

The count $|A_n| = nE_{n-1}$ is due to Kitaev and Mansour [2] and is explicitly restated and attributed in Ehrenborg, Kitaev and Perry [3], Theorem 7.5. Section 7 of [3] supplies the circular-difference invariant subspace and integral-operator framework; these ingredients are not claimed as new. Zhao [4] studies classical occurrences at arbitrary increasing index positions, so its identical-looking

forbidden triples describe a different problem. OEIS A000708 [5] already records $E_{n+1} - 2E_n$ and the EGF $(1 - 2\cos z)/(1 - \sin z)$; neither the sequence nor that EGF is claimed as new.

The potential contribution is their exact identification with the marked consecutive-321 total in this class, together with the three resulting popularity constants. A targeted failure to locate a directly duplicating theorem does not establish originality or priority. **NOVELTY: INCONCLUSIVE.**

9 Reproducibility and evidence provenance

The supplied v0.4 archive is pinned by SHA256:

`9fca07c3ce4f15bfa7c9022660cab8051a175149291a0c38e605dce1c5adc446`

The original mathematical files and code are preserved byte-for-byte. The v0.4.1 PowerShell launcher applies three path-separator conversions to a Python verifier in memory; it is a Windows execution fix and not a new mathematical version.

During preparation of this publication edition, the v0.4 difference verifier passed (19 frozen v0.3 files, 27 v0.4 manifest entries, two editorial-line changes), the original independent permutation enumeration passed for $3 \le n \le 9$, and the exact rational simplex checker passed for every $3 \le n \le 22$. Larger values printed by the original arithmetic script evaluate the claimed formula rather than independently counting occurrences. Finite computation is evidence, not a replacement for the proof.

References [1], [3], [4] and [5] were inspected in this ChatGPT publication-preparation session on 5 November 2026. Theorem 7.5 in [3] attributes the avoidance count to [2]; [2]'s bibliographic metadata was checked against an author-affiliated institutional record. These checks do not change Claude's firsthand REFERENCES PARTIAL status or imply personal inspection by the human author. The owner-supplied Claude report is preserved verbatim in the evidence archive. It reports independent C enumeration for every $3 \le n \le 13$, analytical re-derivation and symbolic checks; that C source was not supplied or rerun here. The owner's Windows console record reports Python 3.12.10, the pinned ZIP hash, all integrity checks, both original verifiers and the final v0.4.1 reproduction PASS. Its output was checked for consistency, but the Windows machine and original transcript-file bytes were not accessed here. Frozen messages saying Windows was untested or publication was on HOLD describe the earlier archival state. Novelty remains **INCONCLUSIVE**.

References

1. Nathanaël Hassler and Sergey Kirgizov. *Emerging consecutive pattern avoidance.* arXiv:2511.02442v2, 10 October 2026. Table 1 (Class 12), equation (1.1), Section 5. https://arxiv.org/html/2511.02442v2.

2. Sergey Kitaev and Toufik Mansour. *Simultaneous avoidance of generalized patterns.* Ars Combinatoria **75** (2005), 267-288. https://arxiv.org/abs/math/0205182.

3. Richard Ehrenborg, Sergey Kitaev and Peter Perry. *A spectral approach to consecutive pattern-avoiding permutations.* Journal of Combinatorics **2**(3) (2011), 305-353. DOI: 10.4310/JOC.2011.v2.n3.a1. Section 7, especially Lemma 7.1 and Theorem 7.5. Manuscript: https://arxiv.org/abs/1009.2119.

4. Alina F. Y. Zhao. *Pattern Popularity in Multiply Restricted Permutations.* Journal of Integer Sequences **17** (2014), Article 14.10.3. https://cs.uwaterloo.ca/journals/JIS/VOL17/Zhao/zhao3.html.

5. OEIS Foundation Inc. *The On-Line Encyclopedia of Integer Sequences*, A000708. Known Euler-number difference and EGF; accessed 5 November 2026. https://oeis.org/A000708.